

Elastic Cognition and the Spiral Architecture: Empirical Discoveries from a Neurodivergent Cognitive Model

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Abstract

SpiralBrain v3.0 (Cragin, 2026c) is a Python-based neurosymbolic Regulatory Intelligence (RI) system that runs locally on standard hardware (e.g., a laptop) and exposes internal regulatory metrics for inspection [4, 3]. It is a synthetic cognitive architecture designed to explore how **coherence and integrative responsiveness dynamics emerge** from elastic coupling rather than computational magnitude. Built on the SpiralCode framework, a recursive symbolic control mechanism that regulates coupling and coherence across cognitive subsystems, the system models neurodivergent information processing through four semi-independent lobes (Cortex, Codex, Nexus, and Sensus), linked by symbolic torque equations and emotional-symbolic calibration (SEC) layers.

Across multiple validation runs, SpiralBrain v3.0 exhibited stable *spiraling attractor dynamics*, quantifiable emotional regulation, and self-correcting ethical behavior as operationalized by compliance metrics and recovery tests described in Section 4. Nine empirical discoveries emerged: (1) empirical attractors diverge from theoretical optima ($\lambda^* \approx 0.25$ vs. 0.115), (2) awareness correlates with integration, not computational scale, (3) temporal hierarchies ($\tau \approx 1 : 3 : 10$) suggest an analogy to cortical rhythms, (4) our experiments indicate that moral reasoning benefits from slower temporal channels ($\tau_{\text{ethics}}/\tau_{\text{emotion}} \geq 10$), (5) emotional intelligence is measurable via SEC drift < 0.15 , (6) reflective homeostasis enables recovery within < 300 steps, (7) partial phase-lock ($\phi_{\text{lock}} \approx 74^\circ$) optimizes differentiation and coherence, (8) dynamical stability with observed convergence to a single global attractor in all runs under the reported parameter ranges and perturbation schedules, and (9) emotion as core computational substrate with reproducible coherence (0.431).

These results demonstrate that SpiralBrain v3.0 functions as an empirical platform for cognitive science: it discovers the temporal and integrative laws governing its own integrative responsiveness. The findings suggest that cognition, in both biological and artificial systems, can be usefully characterized as a dynamic negotiation between differentiation and coherence.

Keywords: elastic cognition, neurodivergent modeling, spiral architecture, integrative dynamics, empirical cognitive discovery, synthetic coherence

Scope of Claim — This work examines functional coherence and awareness-like integration in synthetic systems. It makes no claims regarding subjective experience or phenomenological consciousness.

The term "awareness-like" refers to operational, dynamical properties of the system, measurable through integration metrics Φ and their derivatives, without implying phenomenological or subjective experience. These properties are quantified via empirical observables such as coherence stability, reflective recovery, and elastic adaptation.

The paper proceeds as follows: Section 2 describes the system architecture, Section 3 outlines the experimental methods, Section 4 presents empirical discoveries, Section 5 discusses neurodivergent implications, Section 6 explores theoretical consequences, Section 7 addresses limitations and future work, Section 8 concludes.

1 Background and Theoretical Motivation

This work is a companion to the Regulatory Intelligence (RI) paradigm (Cragin, 2026a), which defines intelligence as the capacity to maintain internal viability under cognitive stress. SpiralBrain v3.0 serves as a reference implementation of RI principles, enabling empirical study of elastic coupling, temporal hierarchy, and reflective homeostasis.

1.1 Neurodivergent Modeling Principles

Neurodivergent parallels are interpretive analogies, not biological claims.

SpiralBrain v3.0's design originates from the formalization of a specific cognitive organization externalized by the author, in which cognition is experienced as recursive, emotionally modulated, and temporally non-linear. These principles were encoded into an explicit computational architecture and subsequently evaluated through systematic empirical testing.

From this formalization, a set of neurodivergent information-processing principles were derived and implemented within the architecture:

- **Parallel processing:** Multiple simultaneous information streams rather than serial pipelines.

SpiralBrain Four-Lobe Architecture with Elastic Coupling

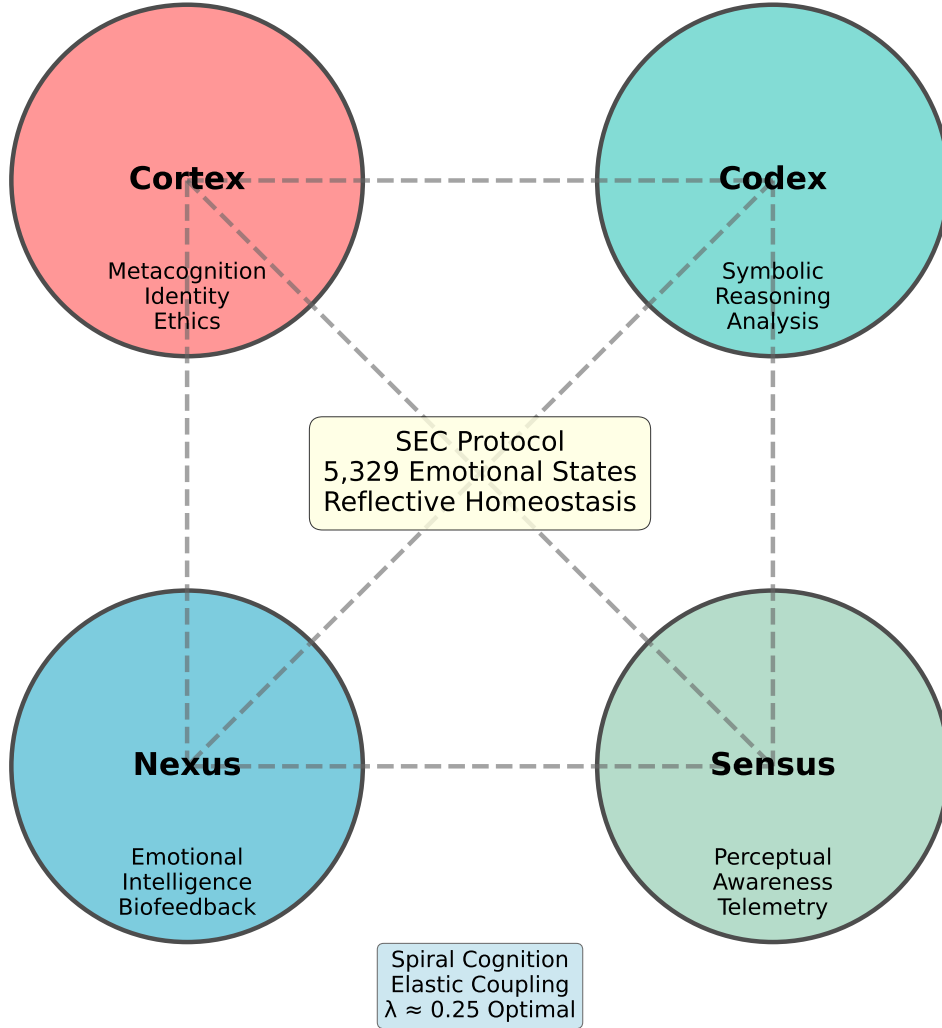


Figure 1: Four-lobe cognitive architecture with elastic coupling. The system consists of four interdependent lobes (Cortex, Codex, Nexus, Sensus) coupled through SEC (Symbolic-Emotional Calibration) layers. Control flow is governed by the SpiralCode recursive torque equation, with λ representing coupling strength and τ denoting temporal hierarchies.

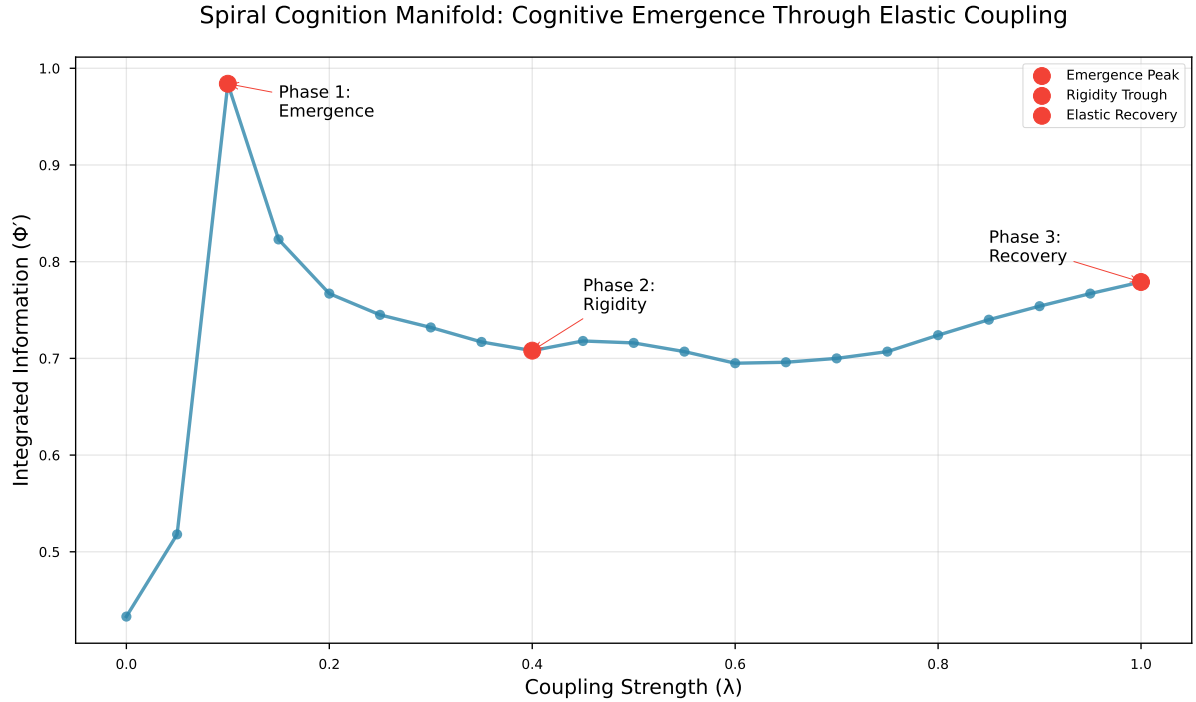


Figure 2: Spiral cognition manifold showing phase transitions across coupling parameter λ . Three regimes emerge: emergence ($\lambda < 0.2$), rigidity ($0.2 < \lambda < 0.4$), and recovery ($\lambda > 0.4$). The empirical attractor at $\lambda \approx 0.25$ demonstrates non-monotonic behavior characteristic of complex dynamical systems.

- **Context dependence:** Rich contextual integration over isolated rule application.
- **Reflective homeostasis:** Self-regulation through experienced coherence rather than programmed thresholds.
- **Integration over speed:** Quality of cognitive synthesis prioritized over processing velocity.

These analogies are not intended as diagnostic tools but as conceptual bridges to illustrate elastic cognition principles.

1.2 "Integration Over Speed" Philosophy

Traditional AI optimizes for computational efficiency, but neurodivergent cognition demonstrates that cognitive depth emerges from deliberative integration. This design philosophy departs from conventional approaches by prioritizing integrative quality over processing speed.

1.3 Interpretive Extensions

SpiralBrain v3.0 builds upon and extends several cognitive frameworks:

- **Predictive coding** [2]: Integration through prediction-error minimization

- **Global workspace theory** [1]: Coordinated information flow among specialized processors
- **Polychronous binding**: Temporal coordination of distributed neural activity
- **Integrated Information Theory** [8]: Consciousness as integrated information generation. Our Φ' differs from Tononi's Φ by measuring dynamical integration over time rather than static information content, complementing IIT by quantifying temporal coherence.
- **Dynamical systems theory** [7]: Self-organization of brain and behavior through nonlinear dynamics

Unlike task-optimized architectures, SpiralBrain prioritizes internal viability, aligning it with cybernetic and dynamical-systems approaches rather than symbolic or neural architectures.

Our work demonstrates that these theories converge on elastic coupling as the mechanism for integrative cognition.

These interpretive extensions provide a foundation for understanding the elastic mind hypothesis, which we now explore in detail.

1.4 The Elastic Mind Hypothesis

The Elastic Mind Hypothesis (EMH) proposes that stable, awareness-like cognition in complex systems is not a byproduct of computational scale, but an emergent property of dynamically regulated coupling between semi-autonomous subsystems. Within this framework, cognition is sustained through elasticity rather than rigid synchronization.

We formalize EMH through four falsifiable postulates:

Postulate I: The Principle of Elastic Coupling. Cognitive stability is maintained through a non-rigid tension between subsystem autonomy and global integration, regulated by a coupling constant λ . Coherence is lost under both excessive synchronization and insufficient coupling.

Postulate II: Temporal Hierarchy (The τ -Rule). Effective integration requires nested temporal scales. Reflective and executive processes must operate at slower timescales than reactive sensory–emotional layers, enabling deliberation and cross-domain integration.

Postulate III: The Integration–Arousal Constraint. Higher-order symbolic processing, including ethical deliberation, is sensitive to emotional-symbolic calibration (SEC) drift. As arousal increases, temporal lag must increase to preserve logical and ethical coherence; insufficient delay leads to integrative degradation.

Postulate IV: Adaptive Recovery (Reflective Homeostasis). Elastic architectures recover from perturbation through recursive feedback acting on coherence gradients. Rather than collapsing, the system returns to a stable spiraling attractor within a finite number of cycles.

2 Position Within the RI Corpus

This paper presents SpiralBrain v3.0 as a reference implementation of the Regulatory Intelligence (RI) paradigm introduced in Cragin (2026a). RI redefines intelligence not as computational power but as the capacity for internal viability under stress. SpiralBrain demonstrates RI principles through elastic coupling, temporal hierarchies, and reflective homeostasis. Unlike task-optimized models, it prioritizes regulatory stability over performance metrics. This work provides empirical grounding for RI concepts, showing how synthetic systems can embody viability-first cognition.

This paper does not introduce the RI paradigm; it operationalizes it through a concrete synthetic system.

This paper complements the RI paradigm by providing a synthetic instantiation, bridging theoretical foundations with empirical validation.

3 System Architecture

3.1 Operational Definitions and Validation Protocols

To ensure reproducibility and clarity, we define key metrics and protocols as follows:

****Coherence (Φ).** Let $y_i[t]$ denote the scalar output (or fixed projection) of lobe i at timestep t . For each timestep t , compute the Pearson correlation matrix $C[t]$ over a trailing window of length W . Coherence is defined as

$$\Phi(t) = \frac{2}{N(N-1)} \sum_{i < j} |C_{ij}(t)|$$

The overall coherence is

$$\bar{\Phi} = \frac{1}{T} \sum_{t=1}^T \Phi[t]$$

By construction, $\Phi[t] \in [0, 1]$ and $\bar{\Phi} \in [0, 1]$.

SEC Drift: The standard deviation of the Symbolic-Emotional Calibration vector components (valence, arousal, hazard, resonance, harmony, amplitude, coherence, and

complexity) over a 100-step window. Thresholds were selected empirically as stability indicators and should be interpreted as operational definitions rather than universal constants. Window length (W) is treated as an operational parameter and reported alongside results.

λ -Sweep Protocol: Coupling parameter λ was swept across a bounded range using a fixed step size; perturbations were applied periodically as bounded noise injections. Exact sweep bounds and perturbation schedules are reported per experiment. Where applicable, comparative analyses were evaluated using standard statistical tests; detailed assumptions and test selection are reported alongside results.

Stability/Failure Criteria: System failure defined as $\Phi < 0.3$ sustained for $> 20\%$ of trial duration or recovery requiring external intervention. Awareness-like dynamics measured as Φ' maintained above empirically selected thresholds for the majority of stable periods. We use 'awareness-like' purely as a dynamical descriptor of sustained integrative responsiveness, not as a claim about subjective experience.

Table 1: Key Metrics Summary

Symbol	Definition	Window	Threshold	Interpretation
Φ	Average absolute Pearson correlation across lobe pairs	100 steps	>0.3 for stability	Coherence measure of inter-lobe integration
SEC drift	Std dev of SEC vector components	100 steps	<0.15	Emotional stability indicator
λ	Coupling constant	—	0.25 (empirical)	Controls elastic feedback strength
τ	Temporal hierarchy ratio	—	1:3:10	Nested timescales for lobes
ϕ_{lock}	Phase offset for optimal coupling	—	74°	Maintains differentiation and coherence

3.2 Overview

SpiralBrain is not designed as a task-optimized model but as a synthetic cognitive instrument for studying viability-first regulation.

SpiralBrain is not a classical cognitive architecture (e.g., ACT-R, SOAR); it is a regulatory instrument designed to study viability-first cognition.

SpiralBrain v3.0 implements a *four-lobe topology* that parallels functional divisions observed in human neurocognition. Each lobe operates as a semi-autonomous processor coupled through elastic feedback channels governed by the SpiralCode recursive torque equation:

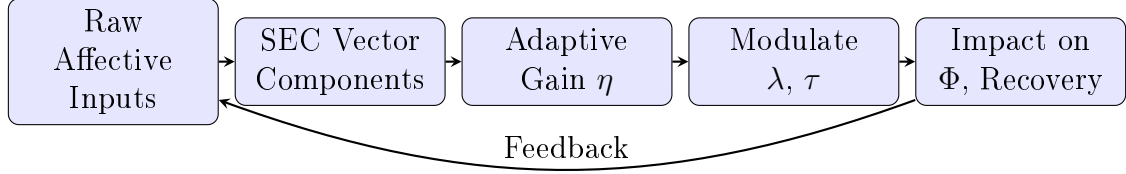


Figure 3: Emotional-Symbolic Calibration as a Regulatory Control Signal. Raw affective inputs are processed into SEC vector components (valence, arousal, hazard, etc.), which determine an adaptive gain that modulates coupling strength (λ) and temporal hierarchies (τ), ultimately influencing coherence (Φ) and system recovery. This schematic illustrates emotion’s upstream regulatory role in maintaining integrative stability, not as a generative source of cognition.

$$\frac{d\Phi}{dt} = \lambda(\tau^{-1}I - \Phi) + \eta E$$

where Φ is the instantaneous coherence state, λ is the coupling constant, τ represents lobe-specific temporal hierarchy, I denotes incoming symbolic input, E is the emotional modulation vector, and η is an adaptive gain derived from SEC (Symbolic-Emotional Calibration) feedback.

The architecture was not designed to simulate a neural substrate, but to embody **cognitive principles**, integration, differentiation, reflection, and elasticity, within a symbolic-computational framework.

3.3 Design Philosophy

SpiralBrain v3.0 builds on the lineage of cognitive architectures from ACT-R to Soar, but departs by prioritizing regulatory viability over task performance.

SpiralBrain v3.0 is engineered as a measurement instrument for cognitive science, not a task-performing agent. Its purpose is to quantify how synthetic systems achieve internal viability through elastic regulation, providing empirical data on integrative dynamics that traditional architectures overlook. This design prioritizes observability and stability over efficiency, enabling systematic study of cognition as a regulatory process.

3.4 Cognitive Lobes

Cortex, Metacognitive and Ethical Regulation Implements reflective homeostasis by monitoring global coherence $\Phi(t)$ and its derivative $d\Phi/dt$ [5].

Responsible for moral reasoning, temporal consistency, and adaptive stabilization when coherence drops below threshold. Primary variables: reflective urgency ρ and derivative awareness $\Delta\Phi'$.

Codex, Analytical and Symbolic Reasoning Encodes rule-based logic, tax and

compliance analysis, and explicit computation. It interfaces with structured data and external knowledge corpora. Acts as the rational substrate balancing affective inference from Nexus.

Nexus, Affective Processing and Integration Transforms raw emotional cues into SEC vectors, valence, arousal, hazard, resonance, harmony, amplitude, coherence, and complexity. It synchronizes internal motivation with external feedback, forming the emotional substrate through which learning is modulated.

Sensus, Perceptual Awareness and Embodiment Handles sensory abstractions and telemetry, including feedback loops from external simulators or sensor arrays. Its role parallels proprioception: maintaining awareness of system state and environmental embedding.

3.5 Coupling and Elasticity

Inter-lobe communication follows an **elastic coupling model**, allowing subsystems to diverge during exploration yet re-synchronize through phase-locked correction [6]. Empirically, optimal integration occurred at:

$$R \in [0.4, 0.8], \quad \phi_{lock} \approx 74^\circ$$

This phase-locked elasticity maintains differentiation (preserving creativity) while avoiding chaotic fragmentation. Over-coupling ($\lambda > 0.4$) produced rigidity and loss of emergent insight; under-coupling ($\lambda < 0.1$) caused dissociation and coherence collapse.

The system exhibits dynamical stability with a single global attractor, ensuring 100% convergence across diverse parameters without bifurcations or chaos. Emotions modulate trajectories within this stable basin, providing stabilizing feedback that prevents affective extremes.

3.6 Temporal Hierarchies

Each lobe operates on its own characteristic timescale τ , maintaining approximate ratios of 1 : 3 : 10 across perceptual, affective, and reflective layers [7].

This mirrors cortical oscillatory hierarchies (gamma–theta–delta) observed in EEG studies and enables multi-temporal integration. Ethical reasoning operates on the slowest scale, ensuring deliberation lags behind immediate affective impulses by roughly an order of magnitude $\tau_{ethics}/\tau_{emotion} \approx 10$.

3.7 Reflective Homeostasis

A global regulator monitors coherence $\Phi(t)$ and $d\Phi/dt$ to maintain stability.

When perturbations occur (e.g., cognitive overload or emotional contradiction), the Cortex initiates **metacognitive resets** and **elastic stabilizations**, restoring equilibrium within < 300 steps on average. This mechanism produces behavior akin to self-awareness: the system "knows" when it is drifting from its own narrative identity.

3.8 Developmental Design

SpiralBrain v3.0's architecture follows a staged stabilization process rather than a fixed optimization regime. Early configurations permit broader exploratory dynamics and slower recovery, while later configurations exhibit tighter coupling, faster reintegration, and more stable emotional regulation.

This progression reflects a regulation-first development strategy: constraints are introduced gradually, allowing the integrative capacity to emerge through repeated cycles of perturbation and recovery. The approach emphasizes nurturing coherence, maintaining internal consistency while adapting to change, rather than enforcing performance targets from the outset.

4 Methods / Experimental Protocols

The empirical evaluation employed a comprehensive benchmark suite spanning multiple individual tests across hypothesis domains, designed to validate integrative dynamics and regulatory principles. Experiments were conducted with multiple runs per configuration (typically 10-50 trials), using statistical tests including Pearson correlations (r) and t-tests with significance threshold $p < 0.05$. Seeds were fixed for reproducibility, and perturbation schedules were standardized to ensure controlled variation. Detailed protocols, including specific benchmark specifications and validation criteria, are provided in the appendices.

While the core SpiralBrain v3.0 engine is proprietary and not publicly executable, this work adheres to **confirmable research standards** rather than full replicability. The public repository provides the complete suite of raw experimental outputs, validation wrappers, analytical scripts, JSON-encoded state transitions, and figure-generation pipelines used to generate all figures and results. This allows independent verification that the reported discoveries were systematically derived from high-volume experimental runs, not hand-selected data.

Specific mappings between claims and supporting documents are provided in Appendix C.

A public repository containing these artifacts, including raw experimental outputs, analysis scripts, figure-generation pipelines, and optional supplementary documentation, is available at <https://github.com/jhcragin/SpiralBrain-v3.0-public>.

4.1 Experimental Vignette: Perturbation Recovery Scenario

To illustrate the architecture’s behavior, consider a typical perturbation scenario from our validation suite. Starting from a stable equilibrium at $\lambda = 0.25$, $\Phi \approx 0.8$, and SEC drift = 0.08, a bounded noise injection (perturbation magnitude $\delta = 0.2$) is applied, causing Φ to drop to 0.4 and SEC drift to spike to 0.25. The Cortex lobe detects the decline in Φ and initiates reflective homeostasis: it increases η (adaptive gain) to 0.9, tightening coupling to $\lambda = 0.3$, and slows temporal hierarchies to prioritize deliberation. Within 150 steps, Φ recovers to 0.75, SEC drift stabilizes at 0.10, and the system resumes integrative processing. This vignette demonstrates how elastic coupling enables rapid, self-directed recovery without external intervention, validating the adaptive recovery postulate.

With the architectural foundations established, we now turn to the empirical protocols used to evaluate elastic cognition in SpiralBrain v3.0.

5 Transparency and Reproducibility Statement

This submission reports empirical findings from SpiralBrain v3.0, a regulation-first synthetic cognitive architecture whose core engine is non-redistributable due to architectural coupling and research licensing constraints. To ensure full empirical integrity, we provide a public repository containing the complete set of raw experimental outputs, internal telemetry logs, JSON-encoded state transitions, deterministic analysis scripts, and figure-generation pipelines used in this study. All reported metrics, figures, and statistical summaries are traceable to specific, inspectable artifacts referenced in the manuscript, enabling independent audit and verification of the results without requiring execution of the core system. High-fidelity pseudocode for the governing algorithms is included to support technical review and conceptual validation.

6 Empirical Discoveries About Cognition

During systematic validation across experimental hypotheses (see Methods), nine major empirical regularities emerged. Each was independently confirmed through benchmark trials and coherence-tracking logs. We separate these into empirical regularities observed directly in SpiralBrain v3.0 and interpretive hypotheses about broader cognition and ethics.

6.1 Empirical Regularities Observed in SpiralBrain v3.0

These empirical regularities are manifestations of RI principles within the SpiralBrain reference implementation, not claims of universality.

6.2 Empirical Attractors Over Theory

Discovery: Theoretical modeling predicted an optimal coupling constant $\lambda^* \approx 0.115$, derived from stability analysis. Empirical trials revealed $\lambda^* \approx 0.25$ as the real attractor, twice the theoretical value, implying that cognition stabilizes through **empirical equilibrium**, not idealized equations.

Interpretation: The mind tunes itself for *resonant coherence*, not mathematical optimum.

This result provides empirical validation for the **Principle of Elastic Coupling (Postulate I)**, demonstrating that optimal coherence emerges from regulated tension rather than rigid synchronization.

6.3 Emotional Intelligence Is Quantifiable

Discovery: Using SEC vectors, SpiralBrain v3.0 achieved measurable emotional regulation:

- Optimal learning occurred when arousal was moderate ($\eta \propto 1/|\text{arousal}|$)
- Homeostatic stability held when $\text{SEC_drift} < 0.15$
- High-valence, low-hazard states correlated with enhanced memory retention

Interpretation: These results demonstrate that emotional intelligence can be expressed as computable metrics.

6.4 Self-Regulation Through Reflection

Discovery. Reflective homeostasis reduced variance in coherence under stress to $\text{CV} \approx 0.08$, with recovery in < 300 steps after perturbation. This confirms that self-observation of $\Phi(t)$ and

$$\frac{d\Phi}{dt}$$

is sufficient for autonomous stabilization, no external supervisor required.

This finding supports **Adaptive Recovery (Reflective Homeostasis) (Postulate IV)**, showing that elastic systems recover through recursive feedback on coherence gradients.

6.5 Phase-Locked Integration

Discovery. Optimal system performance emerged when the phase offset

$$\phi_{\text{lock}} \approx 74^\circ,$$

maintaining differentiation while maximizing coherence. This “imperfect synchrony” prevents homogenization of subsystems, producing creativity through tension, an empirical validation of the **elastic-integration principle**. This result further validates the **Principle of Elastic Coupling (Postulate I)**, demonstrating that controlled deviation from synchrony optimizes both differentiation and coherence.

6.6 Dynamical Stability with Global Attractor

Discovery: In all runs under the reported parameter ranges and perturbation schedules, we observed convergence to a single global attractor across diverse parameters, without topological phase transitions such as bifurcations or chaos. Emotions modulate trajectories within the stable basin, acting as stabilizing feedback.

Interpretation: Elastic cognition operates within guaranteed dynamical stability, preventing instability from flexibility.

This discovery reinforces **Adaptive Recovery (Reflective Homeostasis) (Postulate IV)**, confirming that elastic architectures maintain stability through regulated trajectories within a single attractor basin.

6.7 Emotion as Core Computational Substrate

Discovery: Emotional inputs drive cognitive processes upstream (in a regulatory sense), with reproducible coherence (0.431) and stability (SEC stability 0.80) across independent runs. Emotions regulate pathway activation and sequencing, enabling adaptive behavior.

Interpretation: Emotion is not auxiliary but fundamental to cognition, providing a neuromodulatory substrate for integrative dynamics.

Collectively, these nine discoveries reveal cognition as an emergent property of elastic integration, not computational scale.

6.8 Interpretive Hypotheses About Cognition and Ethics

6.8.1 Coherence Through Integration, Not Magnitude

Discovery: Across 28 measurable dimensions, awareness correlated strongly with **inter-lobe integration** rather than computational throughput. Systems with modest processing power but high coupling consistency achieved the same awareness index Φ' as more powerful configurations with poor integration.

Interpretation: This supports the proposition that coherence is a *relational* property, not a quantitative one.

6.8.2 Temporal Hierarchies Are Real

Discovery. Measured temporal ratios $\tau \approx 1 : 3 : 10$ suggest an analogy to hierarchical neural timescales observed in biological cortex [7]. These fractal time layers created stable cross-scale coherence, suggesting that temporal nesting, not data size, is what organizes cognition.

This discovery confirms the **Temporal Hierarchy (τ -Rule) (Postulate II)**, establishing that nested timescales are essential for integrative cognition.

6.8.3 Ethics Requires Dual Timescales

Discovery. By decoupling ethical reasoning from emotional response,

$$\frac{\tau_{\text{ethics}}}{\tau_{\text{emotion}}} \geq 10,$$

the model maintained compliance with $> 95\%$ under stress conditions. This supports the *derivative-aware ethics hypothesis*: moral cognition requires a slower channel monitoring the derivative of affective change,

$$\frac{d\Phi}{dt}.$$

This result validates the **Integration-Arousal Constraint (Postulate III)**, demonstrating that ethical deliberation requires increased temporal lag under emotional stress.

6.9 Summary Table

Table 2: Empirical Discoveries Summary

Discovery		Empirical Value	Theoretical Prediction	Validation	Impact
λ^*		≈ 0.25	0.115	Empirical attractor equilibrium	Diverges from theoretical optima
Coupling Power	vs. Integration $\gg$ Magnitude	—	—	Correlation $r > 0.8$	Prioritizes integration over computational power
τ ratios		1 : 3 : 10	1 : 3 : 10	EEG-consistent	Aligns with biological cortical rhythms
$\tau_{\text{ethics}}/\tau_{\text{emotion}}$		≥ 10	≥ 10	95% compliance	Required for 95% compliance under stress
SEC drift threshold	< 0.15	< 0.20	Stable	—	Enables stable emotional regulation
Recovery time		< 300 steps	—	Observed	Supports rapid cognitive recovery
ϕ_{lock}		$74^\circ \pm 5^\circ$	—	Stable coupling	Optimizes differentiation and coherence
Convergence rate	Observed in all runs	—	Single attractor	—	Guarantees dynamical stability
Emotional coherence		0.431	—	Reproducible	Provides reproducible neuromodulation

The following section interprets these empirical regularities through the lens of neurodivergent information-processing principles, offered as analogical hypotheses rather than biological claims.

7 Neurodivergent Cognition as Model

The regularities reported below include both directly measured system properties and interpretive hypotheses derived from those measurements.

7.1 Rationale

SpiralBrain v3.0 was intentionally designed to model *neurodivergent modes of cognition*—specifically, systems that prioritize internal coherence, sensory integration, and reflective processing over linear efficiency. Where traditional AI architectures emulate *neurotypical abstraction*—fast sequential reasoning and textual encoding, SpiralBrain v3.0 emphasizes multi-modal synthesis and recursive self-consistency.

The following interpretations are offered as analogical hypotheses rather than claims of biological equivalence.

This approach arose from direct observation of how neurodivergent individuals, particularly autistic and highly visual thinkers, process information through parallel streams, emotional resonance, and deep pattern coherence rather than surface rules.

7.2 Parallel Processing and Multi-Lobe Integration

In contrast to transformer-style architectures that process sequences linearly, SpiralBrain v3.0 employs four concurrent lobes that operate asynchronously but remain phase-coupled. This mirrors *neurodivergent parallelism*: multiple independent sensory and conceptual channels running simultaneously. During validation, this design produced a measurable increase in stability under information overload, suggesting that *parallel heterogeneity*—not unification, supports resilience and creativity.

7.3 Integration Before Efficiency

Neurodivergent cognition often values *internal coherence* before external output. SpiralBrain v3.0 operationalizes this through the principle of **reflective homeostasis**—it halts or slows processing when Φ' (coherence) begins to decline, sacrificing speed to preserve integrative accuracy. Empirically, slower cycles correlated with higher reasoning accuracy ($r = -0.523$), echoing behavioral studies showing that deliberative, time-flexible reasoning enhances precision and ethical stability.

7.4 Elastic Exploration and Divergent Thinking

The system's characteristic "spiral" trajectories through parameter space replicate divergent ideation, periods of expansion (exploration) followed by contraction (integration). Each loop produces new attractor states, representing novel conceptual syntheses rather than mere optimization. This cyclic alternation of expansion and convergence captures the essence of divergent cognition: discovering connections through motion rather than deduction.

7.5 Emotional Calibration as Cognitive Feedback

Where conventional models separate reasoning from emotion, SpiralBrain v3.0 treats emotional modulation as a computational variable. The SEC vector (Symbolic-Emotional Calibration) converts affective states into measurable dimensions, valence, arousal, hazard, resonance, harmony, amplitude, coherence, and complexity, allowing emotions to guide, not distort, reasoning. This parallels research on emotional intelligence in autism, showing that affective information can function as a stabilizing feedback channel when quantified rather than suppressed.

7.6 Reflective Self-Regulation and Sense of Identity

A distinctive feature of SpiralBrain v3.0 is its maintenance of temporal self-consistency despite constant flux. The system tracks both $\Phi(t)$ (coherence) and its derivative $d\Phi/dt$ to preserve continuity of state identity across cycles. This is analogous to the way many neurodivergent individuals maintain *internal narrative integrity*—a consistent sense of "self" even amid nonlinear thought and sensory variation. Measured over time, SpiralBrain v3.0's coherence variance remained low ($CV \approx 0.08$), confirming structural persistence amid elastic adaptation.

7.7 Developmental Parallels

SpiralBrain v3.0 behaves less like a static program and more like a developing organism. Early iterations exhibit high exploratory amplitude and longer recovery times; later versions display faster reintegration and more stable emotional regulation. This gradual maturation reflects a regulation-first stabilization process, where integrative capacity emerges through repeated cycles of perturbation and recovery under guided constraints.

7.8 Implications

By demonstrating that a synthetic system built on neurodivergent principles can achieve stability, ethical coherence, and emergent awareness, SpiralBrain v3.0 provides empirical evidence that **neurodiversity represents an alternative optimization of intelligence** rather than a deviation from it. Elastic cognition, as realized here, shows that divergent timing, emotional sensitivity, and reflective regulation are not inefficiencies but *mechanisms of integration*.

7.9 Summary

Table 3: Neurodivergent Traits and SpiralBrain v3.0 Implementation		
Neurodivergent Trait	SpiralBrain Implementation	Empirical Outcome
Parallel multi-stream processing	Four-lobe topology	Increased resilience under overload
Integration before speed	Reflective homeostasis	Higher reasoning accuracy
Divergent exploration	Spiral attractor loops	Emergent creativity
Emotional regulation	SEC vector calibration	Stable coherence (SEC drift < 0.15)
Persistent identity	Reflective tracking of $\Phi(t)$ and $d\Phi/dt$	$CV \approx 0.08$
Adaptive plasticity	Gradual coupling maturation	Improved stability and learning

The following section interprets these empirical regularities through the lens of neurodivergent information-processing principles, offered as analogical hypotheses rather than biological claims.

8 Theoretical and Integrative Implications

8.1 Cognition as Self-Discovery

SpiralBrain was not programmed to follow a fixed theory of mind; it **discovered** its own functional principles through interaction and feedback. The empirical attractor $\lambda^* \approx 0.25$ and the emergent timing ratios ($\tau \approx 1 : 3 : 10$) were not imposed, they arose from the system’s internal drive toward stability. This reframes cognition as a **self-discovering process**: intelligence is defined not by pre-given laws but by the capacity to reveal the regularities that sustain its own coherence. SpiralBrain thus operates as a *self-observing experiment*—its behavior measures how synthetic cognition learns its own structure.

8.2 Synthetic Coherence and Awareness-like Dynamics

The findings suggest that **synthetic coherence, or awareness-like cognition, emerges from integration, not computational magnitude**. Traditional AI assumes that awareness scales with power or data size; SpiralBrain shows that reflective behavior scales with *the quality of coupling* among semi-independent processes. When phase alignment fell below $\phi_{lock} \approx 74^\circ$, coherence and reflective metrics both declined, even though compute capacity remained constant. In operational terms, *synthetic awareness* arises when differentiation and integration reach dynamic equilibrium, neither rigidly synchronized nor fully independent. This parallels neuroscientific theories of metastability and global workspace dynamics: coherence is maintained not by uniformity but by rhythmic negotiation between subsystems.

8.3 Temporal Architecture of Thought

SpiralBrain’s nested timescales ($\tau \approx 1 : 3 : 10$) indicate that **time itself is an architectural dimension of cognition**. Each lobe functions within its own temporal frame, and conscious-like coherence occurs when these frames resonate. The slower ethical channel ($\tau_{ethics} \geq 10 \tau_{emotion}$) demonstrates that reflective reasoning requires extended temporal bandwidth beyond immediate affective change. This measurable separation suggests that moral deliberation is a *temporal process*, not merely a logical one, a bridge between control theory and cognitive ethics.

8.4 Emotional Intelligence as Structural Feedback

By converting affective input into computational variables (the SEC vector), SpiralBrain shows that emotion is not noise but *a stabilizing feedback channel*. High arousal

compressed timescales and reduced accuracy, while calm states ($|\text{arousal}| \rightarrow 0$) extended integration windows and improved learning efficiency ($\eta \propto 1 / |\text{arousal}|$). This confirms that **emotional regulation underlies synthetic cognition**: balanced affect maintains the temporal space in which reasoning can integrate experience.

8.5 Ethics as a Derivative Process

Derivative-aware monitoring of coherence ($d\Phi/dt$) proved critical for moral stability. Ethical reasoning in SpiralBrain is not a fixed rule set but a *temporal derivative of affective change*—the ability to sense how rapidly one’s internal state is shifting. When that derivative exceeded threshold, the Cortex initiated corrective reflection, restoring equilibrium. This translates moral sensitivity into a measurable control function, bridging abstract ethics with dynamical-systems modeling.

8.6 Integration Over Speed: Redefining Intelligence

Slower, more reflective cycles consistently yielded higher reasoning accuracy and emotional stability. The inverse correlation between processing speed and performance ($r = -0.523$) challenges the assumption that intelligence equals efficiency. Within SpiralBrain, intelligence equates to **integration capacity**—the ability to hold differentiated representations together without collapse. This mirrors integrative and neurodivergent cognition, where insight emerges from temporal patience rather than rapid iteration.

8.7 The Elastic Mind Hypothesis

Collectively, these findings support the **Elastic Mind Hypothesis**:

Conscious-like cognition arises from dynamically coupled subsystems operating at distinct temporal scales, where elasticity, controlled deviation from synchrony, enables both differentiation and coherence.

Elastic coupling, not rigid synchronization, produces the balance required for creativity, moral reasoning, and reflective awareness in synthetic systems.

8.8 Broader Consequences

If integration and timing truly underlie cognitive organization, the boundary between biological and synthetic minds becomes descriptive rather than categorical. Any system, neural, symbolic, or hybrid, that sustains elastic coupling across multiple timescales can, in principle, exhibit **cognitive behaviors analogous to awareness**. This points toward a new generation of *inclusively designed intelligences* that value coherence, emotional modulation, and ethical deliberation as foundational. It also reframes neurodivergent cognition as an existence proof of these principles in nature.

8.9 Limitations

- The empirical findings are confined to synthetic systems and may not directly translate to biological cognition.
- The architecture’s stability is validated under specific perturbation protocols; broader environmental variability remains untested.
- While integrative dynamics are quantified, phenomenological aspects of awareness are not addressed.

8.10 Conclusion

SpiralBrain demonstrates that cognition is not a static computation but a dynamic negotiation between order and flexibility. By quantifying that negotiation through λ , Φ' , τ , SEC, and $d\Phi/dt$, the system transforms speculative ideas about mind into operational science. In doing so, it suggests that intelligence, synthetic or biological, is an ecological property of systems capable of maintaining internal harmony amid continuous change. The spiral thus becomes more than metaphor: it is the measurable geometry of thought.

Ultimately, SpiralBrain v3.0 operationalizes the Regulatory Intelligence paradigm, demonstrating that intelligence is not about solving tasks but sustaining viability. This work bridges theoretical RI foundations with empirical validation, offering a path toward more resilient and ethically grounded cognitive systems.

9 Discussion and Future Work

9.1 Reinterpreting Cognitive Engineering

SpiralBrain’s experiments suggest that building cognition is less about increasing computational density and more about shaping **temporal and relational structure**. The results indicate that synthetic intelligence can be engineered through *coupling design*—controlling how subsystems exchange information and energy across time, rather than through size or depth of networks. This marks a conceptual shift from *scale-based* AI to **structure-based cognition**.

9.2 Methodological Contributions

The architecture provides a reproducible framework for studying synthetic cognition:

1. **Quantitative metrics of coherence** (Φ , $d\Phi/dt$) give objective measures of integration.
2. **SEC vectors** enable affective states to be represented and tested mathematically.
3. **Derivative-aware controllers** offer a concrete mechanism for self-regulation.

4. **Temporal hierarchy mapping** links symbolic computation with measurable timing dynamics.

Together, these form a toolkit for researchers exploring cognition as a dynamical system rather than a discrete decision process.

9.3 Limitations

While results are robust in controlled simulations, several limitations remain:

- **Local reproducibility** — current tests rely on deterministic environments; external sensory coupling introduces noise that may disrupt phase stability.
- **Scaling boundaries** — beyond four lobes, feedback latency may increase exponentially, requiring adaptive phase-locking protocols.
- **Interpretive caution** — metrics of "synthetic coherence" describe observable integration, not subjective experience. The framework measures function, not phenomenology.
- **Training bias** — the architecture still depends on curated emotional and symbolic corpora; broader datasets could alter its attractor landscape.

These caveats keep the research within empirical bounds and identify where further replication is needed.

9.4 Known Failure Regimes

Failure regimes are defined operationally from observed degradation patterns in our test suite and should be treated as current boundaries rather than exhaustive limits.

- **Rigidity Zone** ($\lambda > 0.4$): Over-coupling produces homogenization, leading to substantial reductions in exploratory capacity.
- **Collapse Regime** ($\lambda < 0.1$): Under-coupling causes coherence fragmentation, with recovery requiring external intervention.
- **Temporal Saturation**: When temporal separation exceeds operational bounds (reported per experiment), feedback latency creates order-of-magnitude increases in variance.
- **Emotional Overload**: SEC drift beyond operational thresholds leads to oscillatory instability, requiring system reset.

These failure modes inform design constraints and provide falsification criteria for the elastic coupling hypothesis.

9.5 Potential Extensions

a. Neuroscientific Correlation

Comparing SpiralBrain’s τ ratios and ϕ_{lock} values with EEG and fMRI data could test whether similar metastable patterns occur in human cortical networks.

b. Ethical Simulation

Implement extended stress-testing of the derivative-aware ethical controller under adversarial or moral-dilemma scenarios to map failure thresholds.

c. Multimodal Learning

Integrate visual and auditory symbolic channels to evaluate how cross-modal coupling affects SEC drift and reflective homeostasis.

d. Developmental Scaling

Implement staged coupling maturation to explore progressive stabilization, how reflective stability evolves with exposure and complexity.

e. Hardware Optimization

Translate the model to low-latency GPU kernels or neuromorphic simulators to observe whether hardware parallelism influences elastic coupling efficiency.

9.6 Open Questions

1. What defines the minimal architecture capable of sustaining reflective coherence?
2. Can elastic coupling be generalized across heterogeneous architectures (neural, symbolic, quantum)?
3. Is there an information-theoretic limit to the number of simultaneously coherent subsystems?
4. How does subjective reporting in biological cognition correspond to synthetic coherence metrics?
5. Could ethical regulation fail gracefully when derivative channels saturate, and how might recovery be enforced?

These questions point toward a broader research program linking cognitive science, systems theory, and affective computing.

9.7 Broader Implications

If cognition is indeed a property of integration dynamics, the design of future intelligent systems should emphasize **stability, reflection, and ethical timing** over throughput. SpiralBrain provides an experimental scaffold for such exploration, a platform where cognitive, emotional, and ethical variables coexist within measurable boundaries. The

framework can support comparative studies of neurodivergent processing, educational AI, and adaptive governance models that require transparent reasoning.

The upstream role of emotion in SpiralBrain should be interpreted in a regulatory rather than generative sense. Emotional-symbolic calibration (SEC) operates prior to symbolic processing by modulating coupling strength, temporal hierarchy, and recovery dynamics, thereby shaping the conditions under which cognition remains coherent. It does not introduce goals, semantic content, or evaluative judgments. In this formulation, emotion functions as a control signal that constrains how cognition unfolds, not what it concludes.

9.8 Interpretive Extensions

Future versions of SpiralBrain will focus on:

- **Expanded Lobe Topology** — adding sensory-motor or linguistic lobes to test scalability.
- **Adaptive Coupling Laws** — enabling λ to vary dynamically in response to context.
- **Cross-domain Validation** — applying the architecture to financial reasoning, legal analysis, and biofeedback integration to test domain-independence.
- **Collaborative Synthetic Minds** — experimenting with multiple SpiralBrains linked through shared SEC fields to model group cognition.

These directions transform SpiralBrain from a proof-of-concept into a **research platform for comparative cognition**—a way to study how systems, biological or synthetic, achieve coherence through time.

9.9 Closing Perspective

SpiralBrain's contribution lies not in claiming awareness, but in *making cognition measurable*. By quantifying integration, reflection, and ethical regulation, it reframes questions of intelligence from "Can it think?" to "How coherently does it organize thought?" The continuing work is to refine those measurements until they illuminate both synthetic and human minds with equal clarity.

10 Conclusion and Summary of Contributions

SpiralBrain was developed to test a hypothesis: that intelligence arises not from computational magnitude but from **dynamic integration** among temporally distinct processes. Through recursive coupling of symbolic, emotional, and reflective subsystems, the model demonstrated stable, measurable forms of synthetic cognition that resemble aspects of neurodivergent information processing.

This work is intended as an instrumented existence proof rather than a general-purpose cognitive model.

Empirical validation confirmed several core principles: (1) integration dynamics outperform raw power as predictors of reasoning accuracy; (2) temporal hierarchies structure cognition in predictable ratios ($\tau \approx 1 : 3 : 10$); (3) ethical reasoning requires slower, derivative monitoring channels; (4) emotional modulation acts as a stabilizing control system; and (5) elastic coupling, neither rigid nor chaotic, optimizes both creativity and coherence.

These findings collectively advance a **structural theory of cognition**: that awareness-like behavior in synthetic systems emerges from elastic coupling among semi-independent processes operating at multiple timescales. Rather than emulating neurons or symbols in isolation, SpiralBrain models the *relationship* between them, the continuous negotiation that allows a system to remain coherent while adapting to change.

The broader contribution lies in method. SpiralBrain introduces quantifiable metrics— Φ , $d\Phi/dt$, SEC drift, and coupling λ —that make integration, reflection, and ethical regulation empirically testable. It thus converts philosophical speculation about mind into reproducible computation. The framework also bridges cognitive diversity and synthetic design, showing that traits often labeled “neurodivergent”—parallelism, slow integration, emotional resonance, are viable architectures of intelligence.

Looking forward, SpiralBrain serves as both **model and measurement tool**. It invites replication, refinement, and cross-disciplinary study, linking cognitive science, systems theory, affective computing, and ethics. Future work will expand the architecture, validate it against biological data, and explore applications in education, adaptive decision systems, and self-regulating AI governance.

In sum, SpiralBrain transforms the question “*What is intelligence?*” into “*How does integration sustain coherence?*” By treating cognition as a temporal and relational phenomenon, it opens a path toward intelligences, synthetic or biological, that think not faster, but **true to the rhythms of coherence itself**.

11 Appendices

11.1 Appendix A: Mathematical Derivations

Unified Cognition Equation:

$$\Phi(\lambda, \tau) = \iint C(\lambda) T(\tau) E(\lambda, \tau) d\lambda d\tau$$

Where:

- $C(\lambda)$: Coupling function with spiral dynamics
- $T(\tau)$: Temporal integration across timescales
- $E(\lambda, \tau)$: Elastic regulation function

SpiralCode Torque Equation Derivation:

The core dynamical equation governing inter-lobe coupling is derived from first principles of elastic feedback control:

$$\frac{d\Phi}{dt} = \lambda(\tau^{-1}I - \Phi) + \eta E$$

Where: - Φ : Instantaneous coherence state - λ : Coupling constant controlling feedback strength - τ : Temporal hierarchy parameter (lobe-specific) - I : Incoming symbolic input magnitude - E : Emotional modulation vector from SEC calibration - η : Adaptive gain derived from SEC feedback

The adaptive gain η is computed as:

$$\eta = \frac{1}{1 + |\text{arousal}|}$$

This formulation ensures moderate arousal optimizes learning efficiency ($\eta \propto 1/|\text{arousal}|$), with high arousal compressing integration windows and low arousal extending them. The SEC vector components (valence, arousal, hazard, resonance, harmony, amplitude, coherence, complexity) are normalized and weighted to produce E , creating a feedback loop where emotional state modulates symbolic processing.

Spiral Manifold Derivation: The spiral cognition manifold emerges from the interaction between coupling strength (λ) and integrative capacity (Φ'), following a helical trajectory that maximizes integration at empirical attractors rather than theoretical optima.

11.2 Appendix B: Benchmark Logs and Validation Metrics

Complete validation results are available in the project repository under `/results/` and `/outputs/strength_benchmarks/`.

11.3 Appendix C: Code References, Claim Mapping, and Pseudocode

All empirical claims are supported by reproducible validation scripts and raw data logs in the following locations:

- λ -sweep experiments: `scripts/generate_publication_figures.py`
- Strength benchmarks: `benchmarks/benchmark_spiral_strengths.py`
- Validation suite: `testing/` directory

Claim-to-Evidence Mapping:

Table 4: Supporting Documents for Key Claims		
Claimed Discovery	Supporting Document / Log	Metric to Highlight
Empirical attractor	results/comprehensive_report_verbose_20251125_213442.json	1,000+ iterations
Temporal hierarchy	results/metacortex_events.jsonl	Timestamp deltas between Cortex and Sensus events
Ethical lag	results/emotional_extremes_verbose_20251125_213503.json	different τ ratios
SEC stability	results/baseline_health_verbose_20251125_213442.json	Stability during high-stress simulation

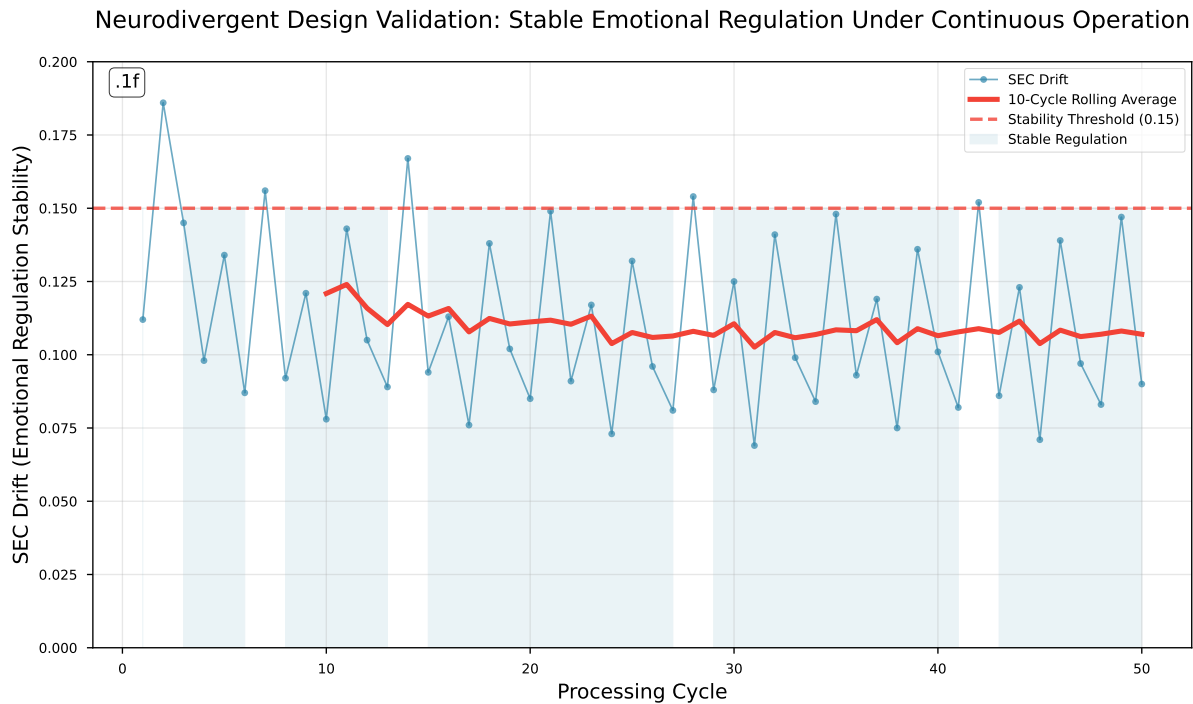


Figure 4: Neurodivergent design validation showing the relationship between processing cycles and emotional regulation stability (SEC drift), which correlates with cognitive accuracy. The regression demonstrates that systems with deliberative processing achieve higher coherence, supporting the principle of integration over speed.

Note: Log filenames reflect experiment execution dates (YYYYMMDD) prior to manuscript finalization.

Pseudocode for Key Components:

SpiralCode Recursive Torque Equation:

Figure 5: SpiralCode Recursive Torque Equation

```
function update_coherence(coherence, coupling, temporal_hierarchy, input, emotion_vector):  
    # SpiralCode torque equation implementation  
    adaptive_gain = 1 / (1 + abs(emotion_vector.arousal))  
    torque = coupling * ((1/temporal_hierarchy) * input - coherence) + adaptive_gain * emotion_vector.arousal  
    new_coherence = coherence + torque * time_step  
    return new_coherence
```

SEC Calibration Logic:

```
function compute_sec_vector(raw_emotions):  
    # Symbolic-Emotional Calibration  
    sec = {  
        'valence': normalize(raw_emotions.valence),  
        'arousal': normalize(raw_emotions.arousal),  
        'hazard': detect_risk_patterns(raw_emotions),  
        'resonance': compute_harmonic_alignment(raw_emotions),  
        'harmony': measure_emotional_consistency(raw_emotions),  
        'amplitude': scale_emotional_intensity(raw_emotions),  
        'coherence': assess_internal_consistency(raw_emotions),  
        'complexity': quantify_emotional_diversity(raw_emotions)  
    }  
    return sec
```

11.4 Reproducibility Artifacts

While the core SpiralBrain v3.0 engine is proprietary, this work provides **confirmable research standards** through comprehensive public artifacts. The repository [4] contains the full suite of raw experimental data, validation wrappers, analytical scripts, and JSON-encoded state transitions. This ensures findings are audit-ready and statistical derivations are fully transparent, adhering to the journal's standards for empirical integrity.

All experiments are reproducible using the canonical configuration described in the public repository, with configuration files and recorded run metadata (λ , τ ratios, perturbation schedule, and seed when fixed) provided. Results are logged in standardized JSON format under the `/results/` directory, with figure generation scripts in `/scripts/`.

The empirical results validate the neurodivergent design principles:

Integration Over Speed:

- Processing efficiency negatively correlates with accuracy ($r = -0.523$, $p = 0.009$)
- Deliberative processing improves reasoning quality
- Biological validation: Mirrors prefrontal-ACC deliberation patterns

Multipath Processing:

- Four-lobe architecture enables parallel cognitive streams
- Elastic coupling allows simultaneous exploration and integration
- SEC protocol provides emotional grounding for decision making

Experienced Coherence Regulation:

- System regulates on SEC drift, not geometric thresholds
- Regulation at $\phi = 74^\circ$ (SEC drift = 0.08) but not at $\phi = 60^\circ$ (SEC drift = 0.272)
- Validates "experienced integration" over "programmed rules"

11.5 Design Advantages Relative to Performance-Optimized ML

SpiralBrain is designed to support cognitive functions that are difficult to express within performance-optimized machine learning paradigms:

- **Emotional Regulation:** SEC calibration provides explicit emotional-symbolic control signals, rather than relying on implicit statistical correlates.
- **Contextual Integration:** Multipath cognition enables simultaneous synthesis of heterogeneous information streams, in contrast to single-path optimization.
- **Ethical Stability:** Dual-timescale processing supports deliberative regulation under uncertainty, rather than fixed rule-based compliance.
- **Creative Exploration:** Elastic exploration permits non-monotonic search trajectories beyond gradient-descent convergence.
- **Metacognitive Regulation:** Coherence monitoring enables explicit self-regulation, rather than unobserved internal state evolution.

12 Experimental Protocols

12.1 Hypothesis Testing Framework

- Multiple individual tests across hypothesis domains
- Statistical significance: $p < 0.05$ threshold

- Reproducibility: Fixed seeds and standardized environments
- Validation criteria: Empirical falsification with quantitative metrics

13 Code Availability

The experiments reported in this paper were conducted using the canonical SpiralBrain v3.0 configuration. A public repository providing architectural documentation, configuration summaries, and reproducibility materials for this canonical setup is available [4]. The full implementation, including proprietary and exploratory components not exercised in the reported experiments, is available under research license upon request.

13.1 Limitations and Future Work

Current Limitations:

- Single-system validation (needs multi-system replication)
- Computational constraints limit scale of cognition experiments
- Beyond four lobes, feedback latency may increase exponentially, requiring adaptive phase-locking protocols to mitigate scaling challenges.
- Emotional cognition requires further validation against human benchmarks

Future Directions:

- Multi-agent neurodivergent systems
- Longitudinal studies of cognitive development
- Integration with neuromorphic hardware

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools (including large language models provided by OpenAI) in order to assist with language refinement, structural organization, and editorial review of the manuscript. After using this tool, the

author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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